A-A-53292
1 December 1987

COMMERCIAL ITEM DESCRIPTION

PROCESSING UNIT, DENTAL RESINS

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a processing unit suitable for processing self-curing dental resins at elevated pressure to eliminate porosity.

Salient Characteristics:

Materials. Materials shall be new and shall conform to the applicable requirements specified herein. Material not specifically specified shall be of high commercial quality throughout, and shall be entirely suitable for the purpose intended.

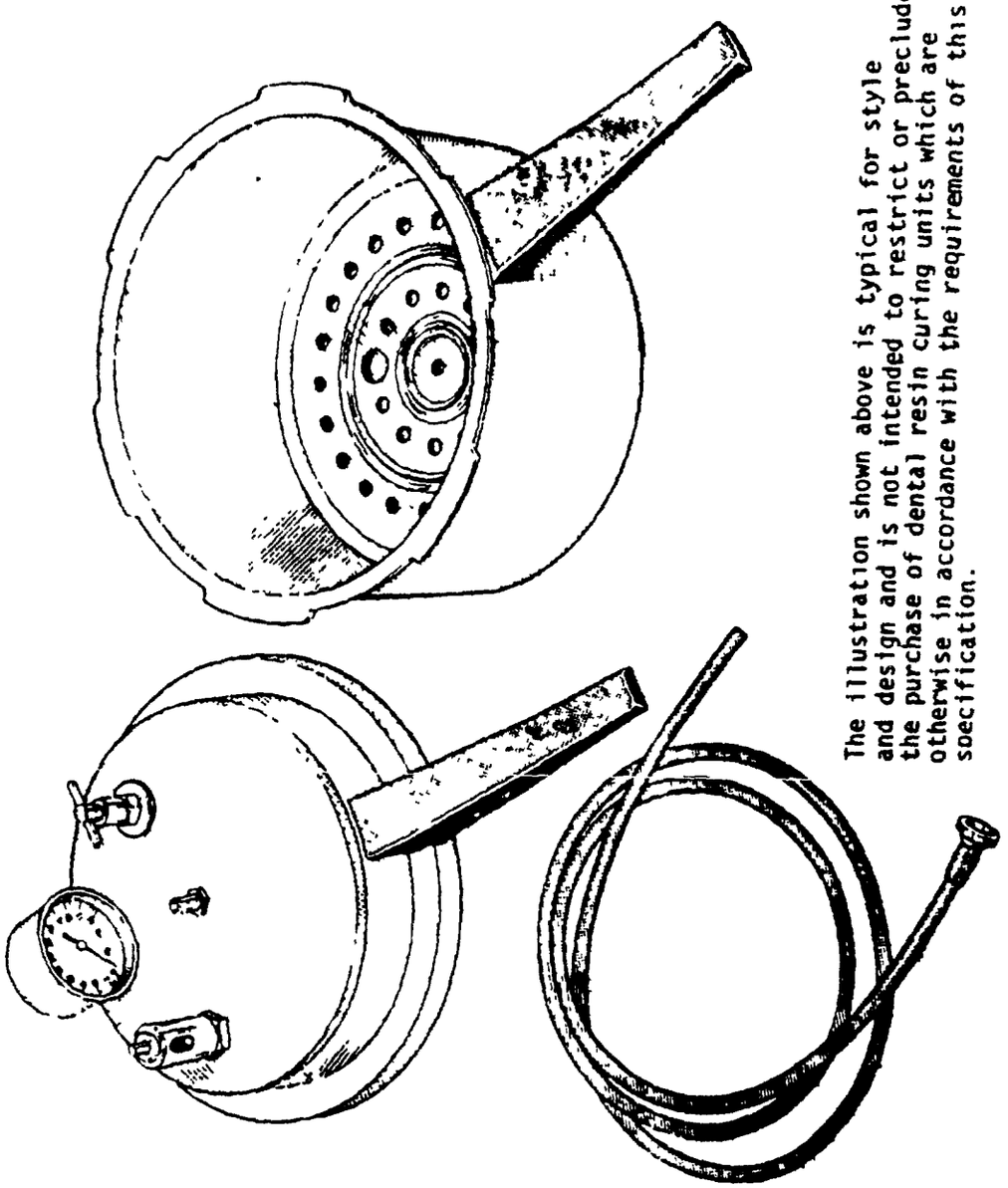
Design and construction. The processing unit shall be a heavy duty unit, as shown in Figure 1, suitable for the processing of self-curing dental resins at 20 ± 5 pounds per square inch (psi), to eliminate porosity. The processing unit shall consist of a six quart, seamless, cylindrical shell having radiused corners; a two part handle; a safety type cover with a cam operating locking device; a pressure gage; an inlet valve; a pressure release valve; a rubber gasket; a spring-loaded safety valve with brass seat with Buna-N disc; and an air hose assembly for compressed air input. The furnishing of a perforated plate for holding the denture to be processed is optional.

Shell. The shell shall be made of one piece, seamless, deep drawn 304 stainless steel, having a minimum thickness of 0.062 inch. The shell shall have a handle fabricated of high temperature phenolic thermosetting plastic having a specific gravity of 1.40 to 1.50. The handle shall be of sturdy design capable of supporting the shell, completely filled with water, without loosening or breaking. The handle shall be securely fastened to the shell in such a manner which will prevent loosening or pulling free. The handle shall mate with the handle on the cover, when the cover is in the locked position. The outside dimensions of the shell shall be 9 ± 1 inches excluding the handle and 16 ± 1 inches including handle.

AMSC N/A FSC 6520

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The illustration shown above is typical for style and design and is not intended to restrict or preclude the purchase of dental resin curing units which are otherwise in accordance with the requirements of this specification.

FIGURE 1. Processing Unit, Dental Resin.

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Cover. The cover shall be rigid and fabricated of seamless Type 201 stainless steel having a minimum thickness of 0.050 inch. The cover shall be formed to hold a removable gasket for positive leak proof seal when the unit is closed. The air inlet valve, the safety valve, the pressure gage and the pressure release valve shall each be fitted and sealed into the cover. A plastic handle shall be fabricated of the same material as specified above, and shall be securely attached to the cover and shall mate with the handle on the shell when the unit is locked into closed position.

Safety valve. The safety valve shall be of rugged construction and suitable for the purpose intended. The valve shall be spring loaded and shall be adjusted to release pressure at 25 to 27 psi, and automatically return the unit to 20 $\pm$ 5 psi. The valve shall have a brass seat with Buna-N disc.

Air inlet valve. The air inlet valve shall be a spring loaded check valve type suitable for the purpose intended.

Pressure release valve. The pressure release valve shall be a screw type quick exhaust valve, with a metal to metal seat of brass construction. The valve shall have a tee handle with a steel crossbar.

Pressure gage. The pressure gage shall be graduated for 0 to 60 pounds per square inch (psi). The minimum unit division shall be 2 psi. The pressure gage shall be of high commercial quality throughout and shall be entirely suitable for the purpose intended.

Gasket. The gasket shall be fabricated of neoprene rubber or chloroprene rubber having a hardness (durometer) of 60 to 70.

Safety disk. A rubber safety disk shall be located underneath the handle of the cover. The disk shall pop into the chamber between the pot handle and cover handle when the pressure reaches 35 plus or minus 5 psi to relieve the pressure. The unit shall be capable of being pressurized again upon reseating of disk by hand. As an alternate, the safety disk shall be located on the cover 3/4 inch from the edge. The disk shall pop into the air when the pressure reaches 35 plus or minus 5 psi to relieve the pressure.

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Air hose assembly. The unit shall be supplied with a 5 feet (minimum) length, 3/16 ± 1/64 inch inside diameter, 3/64 ± 1/64 inch wall thickness rubber or vinyl tubing with a female threaded cap on one end for attachment to the air intake valve.

Finish. The outside surface of the stainless steel (excluding bottom) and the outside surface of the stainless steel cover shall have a polished finish. All surfaces shall be free of corrosion.

Performance. The processing unit shall be capable of maintaining an internal pressure of 20 ± 5 psi. The processing unit shall lose no more than 10 percent of its initial pressure when tested as specified herein. The safety valve shall release the internal pressure when it rises to 25 to 27 psi and then return the internal pressure to 20 ± 5 psi when tested as specified herein. The cover shall not be capable of being easily removed until the internal pressure has decreased below 1 psi when tested as specified herein.

Identification marking. The manufacturer's name or trademark shall be indicated on each unit by stamping on the bottom or by the use of a pressure sensitive label on the upper surface of the plastic handle of the cover.

Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

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Workmanship. The units shall be free from defects which may affect proper functioning, appearance, or may impair their serviceability.

Unit. Each (EA). One processing unit, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one processing unit. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Tests.

Pressure loss. Each closed unit shall be filled with air until the pressure gage reads 20 $\pm$ 5 psi. The unit shall be allowed to stand under pressure for one hour and the gage read to determine pressure loss.

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Safety valve. Each closed unit shall be filled with air under 30 to 32 psi and the safety valve shall be tested to determine if it actuates to release the pressure and return to that pressure specified in performance.

Cover release. Each closed unit shall be filled with air until the pressure gage reads 20 ± 5 psi. The air shall then be released until 1.0 psi remains in the unit. An attempt shall be made to open the unit.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-00-890-1382	PROCESSING UNIT, DENTAL RESINS

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

48503

Project No. 6520-2039

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53292	PROCESSING UNIT, DENTAL RESINS

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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